8. RIDGE REGRESSION : ENERGY

RETURNS PREDICTION

Ridge Regression → introduced by Hoerl and Kennard — biased estim. method for problems where predictors are non orthogonal, meaning correlated. To improve stability and reduce MSE compared with OLS.

OLS can behave poorly in 3 situations

↳ Predictors correlated — e.g. Int. Rates, Bond Yield & Term Spread (Multicollinearity)
$(X^TX)^{-1}$ becomes close to singular. Inverse becomes unstable

↳ Too many predictors (p) relative to data (n)
If $p > n$, OLS can overfit — Ordinary OLS soln. may not even be unique.

↳ High Variance — OLS often has low Bias but potentially high variance
• Ridge delib. adds Bias to reduce Variance.

"A slightly biased model can predict better than an unbiased but unstable model."

OLS minimises RSS

Ridge minimises $RSS + \lambda \sum_{j=1}^{p} \beta_j^2$

Objective of Ridge is :

$$\hat{\beta}_{Ridge} = \arg\min_{\beta} \left[\sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right]$$

$\lambda \sum_{j=1}^{p} \beta_j^2$ is the Penalty. It says we may fit the data but we are punished for making coeff. too large.

In our code we called this α.

$\alpha \approx 3.7276$ (Problem 8)

So, $\lambda \approx 3.7276$

- $\lambda = 0 \Rightarrow$ Ridge becomes OLS
- $\lambda \to \infty \Rightarrow$ Coeff. shrink heavily toward zero, in the limit $\hat{\beta}_j \to 0$.

Ridge uses the penalty $\sum_{j=1}^{p} \beta_j^2 \Rightarrow$ called an L_2 penalty because it is based on the squared length of the coeff. vector.

L_2 norm is :

$$\|\beta\|_2 = \sqrt{\beta_1^2 + \beta_2^2 + \ldots + \beta_p^2}$$

Ridge uses the squared L_2 norm $\|\beta\|_2^2$

So, the Ridge Penalty is $\lambda \|\beta\|_2^2$

Hence it is also called L_2 Regularisation.

Ridge Objective :

$$(Y - X\beta)'(Y - X\beta) + \lambda\beta'\beta$$

where,

$$\beta'\beta = \beta_1^2 + \beta_2^2 + \dots + \beta_p^2$$

Ridge solution is :

$$\hat{\beta}_{Ridge} = \underbrace{(X'X + \lambda I)^{-1} X'Y}$$

on OLS it is $X'X$

Adding λI adds a positive number to the diagonal of $X'X$. This makes the matrix better behaved and easier to invert. Ridge stabilises the regression by strengthening the diagonal of the predictor's covariance matrix.

Deriving the Ridge Solution

Ridge Objective is :

$$L(\beta) = (Y - X\beta)'(Y - X\beta) + \lambda\beta'\beta$$

$$L(\beta) = Y'Y - 2\beta'X'Y + \beta'X'X\beta + \lambda\beta'\beta$$

Diff. w.r.t. β

$$\boxed{\frac{\partial L}{\partial \beta}} = -2X'Y + 2X'X\beta + 2\lambda\beta$$

$$-2X'Y + 2X'X\beta + 2\lambda\beta = 0$$

$$-X'Y + X'X\beta + \lambda\beta = 0$$

$$(X'X + \lambda I)\beta = X'Y$$

$$\boxed{\hat{\beta}_{Ridge}} = (X'X + \lambda I)^{-1} X'Y$$

Unlike LASSO, Ridge usually does not set coefficients exactly equal to zero. So, Ridge shrinks coefficients but usually does not select variables.

Ridge introduces shrinkage → causes Bias to ↑ but Variance ↓

The (Expected Test Error) is:

$$\mathbb{E}[(Y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

Ridge is useful when reduction in variance is larger than the increase in squared Bias.

Ridge is generally biased:

$$\mathbb{E}[\hat{\beta}_{ridge}] \neq \beta$$

But (Prediction Quality) depends on MSE:

$$MSE(\hat{\beta}) = \text{Bias}(\hat{\beta})^2 + \text{Var}(\hat{\beta})$$

The deliberate small amount of bias for a potentially large reduction in Variance can improve OOS prediction.

Correlated Variables / Multicollinearity → Ridge does not need to choose one predictor and discard the other. It spreads the effect more smoothly across correlated predictors.

After (standardisation), → (Feature Scaling)

$$\text{Mean}(X_j) = 0$$

$$\text{Std. Dev.}(X_j) = 1$$

↓

Makes the Ridge Penalty Fair

In Ridge Penalty,

$$\lambda \sum_{j=1}^{p} \beta_j^2 ,$$

see that j starts from 1 and not 0. So, this means that we do not penalise the intercept; usually — intercept does not represent sensitivity to a predictor — penalising it would unnecessarily force the average predict. toward zero. When predictors and target are centred, the intercept is naturally handled separ -ately.

Ridge in Constraint form :

$$\min_{\beta} \text{RSS}$$

subject to:

$$\sum_{j=1}^{p} \beta_j^2 \leq c$$

It says: Fit the data, but the total coeff. size must stay within a budget.

For every suitable value of λ, there is a corresponding constraint level c.

This is useful because it gives a geometric interpretn.

In 2D, suppose we have two coefficients

$$\beta_1 \text{ and } \beta_2$$

The Ridge constraint is :

$$\beta_1^2 + \beta_2^2 \leq c$$

This is a circle.

Ridge Reg. says we can minimise RSS only within this circle. Because the Ridge constraint is smooth and circular, the solution usually lands on the boundary but not exactly on the axis. That is why Ridge shrinks coeff. but usually does not set them exactly to zero.

LASSO uses a diamond shaped constraint —

$$|\beta_1| + |\beta_2| \leq c$$

The diamond has corners on the axes, so LASSO can set coefficients exactly to zero.

Ridge shrinks coefficients while retaining all variables, rather than selecting a subset. Ridge is especially helpful when OLS is trying to fit weak, noisy directions in the data.

Choosing λ : Usually, we choose λ using Cross Validn.

Cross Validn. objective:

$$CV(\lambda) = \frac{1}{K} \sum_{k=1}^{K} Error_k(\lambda)$$

Then choose:

$$\lambda^* = \arg\min_{\lambda} CV(\lambda)$$

For Problem 8: $\lambda^* \approx 3.7276$

The interpretn. of this number depends on:

feature scaling, target scaling, sample size, candidate alpha grid, model implementn., CV design.

Trng. Data → Used to fit the model

Validn. Data → Choose best model setting such as λ.

We choose the λ that gives the lowest validn. error.

Ridge is not trying to minimise trng. error only. It is trying to improve generalisation.

↳ mean how well the model performs on new unseen data.

Ridge coeff. can still indicate direction and relative importance after scaling, but because they are penalised, I would be cautious about interpreting them as unbiased causal effects.

Ridge → Regularised Estimate

Ridge keeps variables; LASSO can remove variables

We usually evaluate prediction using metrics such as:

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{MSE}$$

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{SSR}{SST}$$

Ridge may have slightly worse trng. error but better test error.

Unlike MSE, MAE does not punish large errors as strongly.

MSE / RMSE are more sensitive to large errors.

Regularisation → A method that controls model complexity so that the model does not overfit.

Ridge Reg. is regularised version of Linear Reg.

9. GDP Growth Forecasting

+

10. CO_2 Emissions Intensity

Applied Ridge to Noisier, more complex forecasting problems.

Each predictor may contain only weak signal.

$$\min_{\beta} \left[RSS + \alpha \sum_{j=1}^{p} \beta_j^2 \right]$$

RSS means prediction error on training data.

On Problem 9 : $\alpha = 1873.8174$

Problem 10 : $\alpha = 14166.4327$

A large α means CV preferred stronger shrinkage.

Large α ⇒ Stronger shrinkage ⇒ Lower Variance

⇓

Potentially better test performance

In Problem 10 → In first run, best α was 10,000 — but that was also the max. value in alpha grid.

This is called a **boundary problem**. So we expanded the alpha grid.

We use often, **Log-spaced Alpha Values**. E.g.

0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000

We generally don't use: 1, 2, 3, 4, 5 . . . .

Hyperparameter tuning depends on the search grid — we saw this in Problem 10.

→ A model setting chosen before or during training, not a coeff. estimated directly from the data.
In Ridge, α is a hyperparameter.

Robustness Check → to see if our conclusion survives reasonable changes in the modelling setup. Every number might not remain identical — the main conclusion remains sensible or not.

Alpha Value depends on:

- ↳ how predictors were scaled
- ↳ how target was scaled
- ↳ Train-Test Split
- ↳ CV method
- ↳ Alpha Grid
- ↳ Dataset length
- ↳ Noise level

Ridge Coefficient Path → Track how coefficients change as alpha increases.

The path looks like:

$$\alpha \uparrow \Rightarrow |\beta_j| \downarrow$$

Strong α does not necessarily mean the model is useless. It might mean the model is heavily regularised. Predictions may become closer to a simple benchmark, such as the average value. But that can be sensible if predictors are noisy.

If the data is noisy, an aggressive model may chase false patterns. A heavily regularised model says: I will only move away from the average prediction when the predictors provide reliable evidence.

Historical Mean Benchmark: $\hat{y}_{t+1} = \bar{y}$

Many financial forecasting models fail to beat a simple historical avg. out of sample.

Train on earlier periods
Test on later periods
} Prevents (the look ahead bias)

Macroeconomic and Financial predictors are often (persistent)
↓
Values today are highly related to their values in the past.

Stationarity → Statistical relationship stays broadly stable over time.

E.g. suppose relnship b/w Interest Rates and future growth is stable from 1980 to 2020 — then a model trained on the past may work in the future. But if relationship changes, model may fail. This is called regime change.

↳ E.g. Pre 2008 v. Post 2008
Pre Covid v. Post Covid
Zero Interest Rate v. Rising Rate Period.

Degrees of Freedom → How flexible a model is.

OLS with many predictors — very flexible

Ridge reduces Effective Flexibility — has lower Effective Degrees of Freedom than OLS

↳ How much freedom the model has to bend itself to the training data.

Ridge can fail :

↳ Predictors have no useful signal
↳ Relationship changes over time
↳ Target is mostly noise
↳ Train-Test Split is unlucky
↳ important variables are missing
↳ Model is too linear for true relationship.

If the best hyperparameter lies on the boundary of your search grid, expand the grid before interpreting the result.

Higher Alpha does not automatically mean better performance.

Ridge is a way of making linear forecasting models more stable when the data are noisy and predictors overlap.